

CSA0501
DATABASE MANAGEMENT SYSTEMS
LAB EXPERIMENT

K.NIKHIL

192424366

EXP 5:

AIM:

To view records from tables using SELECT commands with the WHERE clause and pattern matching (LIKE operator).

SELECT with Various Clauses – WHERE, Pattern Matching

ALGORITHM:

1. Start and connect to the csa05_lab database.
2. Extend STUDENT2 with additional columns (DOB, Gender, City) and FACULTY with a JoinDate column, since the questions need this data.
3. Extend COURSE with a HallNo column and add a student–course enrollment table to support course-based queries.
4. Use SELECT column1, column2 FROM table WHERE condition; to answer direct WHERE-based questions (Gender, course, join date, hall allocation).
5. Use SELECT * FROM table WHERE column LIKE '%pattern'; to find names ending with a given substring.
6. Use SELECT * FROM table WHERE column LIKE '%pattern%'; to find names or cities containing a given substring.
7. Use SELECT * FROM table WHERE column NOT LIKE or LENGTH(column) <> n to exclude names of a specific letter count.
8. Use SELECT * FROM table WHERE column LIKE '%pattern%'; for the final "contains" pattern query.
9. Execute all SELECT statements and observe the result sets.
10. Stop.

QUESTIONS:

WHERE:

- 1.The student counselor wanted to display the registration number, student name and date of birth for all the students.
- 2.The controller of examinations wanted to list all the female students.
- 3.List the Students who registered for the “C001” course.

4.Display all faculty details joined before “November 2014”.

5.Display all the courses not allotted to hall ‘H001’.

LIKE:

1.List the students whose name ends with the substring “sh”.

2.Display all students whose name contains the substring “sh”.

3.Find all the students who are located in cities having “Sal” as substring.

4.Display the students whose names do not contain six letters.

5.Find all the students whose names contains “am”.

OUTPUTS:

```
mysql> SELECT RegNo, S1Name, DOB FROM STUDENT2;
+-----+-----+-----+
| RegNo | S1Name | DOB |
+-----+-----+-----+
| 101 | Nikhil | 2004-05-12 |
| 103 | Meena | 2004-08-20 |
| 105 | Divya | 2003-11-02 |
| 106 | Arjun | 2004-01-15 |
| 107 | Sneha | 2004-03-09 |
| 191711342 | Karthik Raj | 2003-07-25 |
+-----+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM STUDENT2 WHERE Gender = 'F';
+-----+-----+-----+-----+-----+-----+
| RegNo | S1Name | Branch | DOB | Gender | City |
+-----+-----+-----+-----+-----+-----+
| 103 | Meena | IT | 2004-08-20 | F | Salem |
| 105 | Divya | IT | 2003-11-02 | F | Chennai |
| 107 | Sneha | CSE | 2004-03-09 | F | Madurai |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT S.RegNo, S.S1Name
-> FROM STUDENT2 S
-> JOIN STUDENT_COURSE SC ON S.RegNo = SC.RegNo
-> WHERE SC.CourseNo = 501;
```

```
+-----+-----+
| RegNo | S1Name |
+-----+-----+
| 101   | Nikhil |
| 103   | Meena  |
+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM COURSE WHERE HallNo <> 'H001';
```

```
+-----+-----+-----+-----+
| CourseNo | CourseName          | SemNo | HallNo |
+-----+-----+-----+-----+
| 502      | DAA                  | 4     | H002   |
| 504      | Computer Networks   | 6     | H003   |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM STUDENT2 WHERE S1Name LIKE '%sh';
```

```
+-----+-----+-----+-----+-----+-----+
| RegNo | S1Name | Branch | DOB          | Gender | City    |
+-----+-----+-----+-----+-----+-----+
| 108   | Niles | CSE    | 2004-02-14  | M      | Chennai |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM STUDENT2 WHERE S1Name LIKE '%sh%';
```

RegNo	S1Name	Branch	DOB	Gender	City
108	Nilesh	CSE	2004-02-14	M	Chennai

1 row in set (0.00 sec)

```
mysql> SELECT * FROM STUDENT2 WHERE City LIKE '%Sal%';
```

RegNo	S1Name	Branch	DOB	Gender	City
103	Meena	IT	2004-08-20	F	Salem
106	Arjun	ECE	2004-01-15	M	Salur

2 rows in set (0.00 sec)

```
mysql> SELECT * FROM STUDENT2 WHERE LENGTH(S1Name) <> 6;
```

RegNo	S1Name	Branch	DOB	Gender	City
103	Meena	IT	2004-08-20	F	Salem
105	Divya	IT	2003-11-02	F	Chennai
106	Arjun	ECE	2004-01-15	M	Salur
107	Sneha	CSE	2004-03-09	F	Madurai
191711342	Karthik Raj	AI & DS	2003-07-25	M	Chennai

```
mysql> SELECT * FROM STUDENT2 WHERE S1Name LIKE '%am%';
```

RegNo	S1Name	Branch	DOB	Gender	City
109	Amitha	IT	2004-06-30	F	Trichy

1 row in set (0.00 sec)

RESULT:

Records from the tables were successfully displayed using SELECT commands with the WHERE clause and pattern matching (LIKE operator) — filtering data based on gender, course registration, join date, hall allocation, and various name/city substring patterns.